

PAPER_382 — UQFF 12-Term Full Spectral Ladder for SGR1745

Session: 104 (Complete Re-Analysis — per-term numeric tabulation undiscovered in PAPER_371)

PAPER_382 — UQFF 12-Term Full Spectral Ladder for SGR1745

Source: grokshare11254865.txt, lines ~2920–2950 **Section:** SGR1745 resonance MUGE computation — all 12 terms with numeric values **Session:** 104 (Complete Re-Analysis — per-term numeric tabulation undiscovered in PAPER_371) **CP4 Class:** UQFF12TermSpectralLadderSGR1745Calculator (CP4 #33)

1. Overview

PAPER_371 documented the 12-term resonance MUGE formula and structure (the co-sum of 12 independent physical mechanisms). It did NOT record the **actual numerical value for each individual term** for any specific system. This paper provides the **first per-term numeric tabulation** for SGR1745 — establishing a 78-order dynamic range spectral ladder.

The discovery: $a_{\text{fluid}\backslash\text{freq}}$ dominates by **75 orders of magnitude** above the weakest term $a_{\text{Aether}\backslash\text{freq}}$. This defines a UQFF term hierarchy for compact objects.

2. System Parameters (SGR1745 Magnetar)

Parameter	Value	Units
Mass M	2.984×10^{30}	kg
Radius r	$1 \times 10^{\blacksquare}$	m
B-field	1×10^{10}	T
B_crit	1×10^{11}	T
Age t	3.799×10^{10}	s
Redshift z	0.0009	—
V_sys	4.189×10^{12}	m^3
v_exp	1×10^3	m/s
f_fluid	$1.269 \times 10^{-1} \blacksquare$	Hz
E_vac,neb	$7.09 \times 10^{-3} \blacksquare$	J/m^3
E_vac,ISM	$7.09 \times 10^{-3} \blacksquare$	J/m^3

3. Complete 12-Term Spectral Ladder

Term 1: Dipole-Phase-Modulation Acceleration (aDPM)

$$a_{\text{DPM}} = F_{\text{DPM}} \cdot f_{\text{DPM}} \cdot E_{\text{vac,neb}} \cdot c \cdot V_{\text{sys}}$$

Where $F_{\text{DPM}} = I \cdot A \cdot (\omega_1 - \omega_2)$ with $I=10^{21}$ A, $A=3.142 \times 10^8 \text{ m}^2$, $\omega_1 - \omega_2 = 10^{12} - 9.99 \times 10^{11} = 10^9 \text{ rad/s}$:

$$\boxed{a_{\text{DPM}} = 3.545 \times 10^{-42} \text{ m/s}^2}$$

Term 2: THz Frequency Coupling (a_{THz})

$$a_{\text{THz}} = f_{\text{THz}} \cdot \frac{E_{\text{vac,neb}} \cdot v_{\text{exp}} \cdot a_{\text{DPM}}}{E_{\text{vac,ISM}} \cdot c}$$

With $f_{\text{THz}} = 10^{12}$ Hz:

$$\boxed{a_{\text{THz}} = 1.182 \times 10^{-33} \text{ m/s}^2}$$

Term 3: Vacuum Energy Differential ($a_{\text{vac_diff}}$)

$$a_{\text{vac_diff}} = \frac{\Delta E_{\text{vac}} \cdot v_{\text{exp}}^2 \cdot a_{\text{DPM}}}{E_{\text{vac,neb}} \cdot c^2}$$

$$\boxed{a_{\text{vac_diff}} = 3.545 \times 10^{-53} \text{ m/s}^2}$$

Term 4: Super-Frequency Coupling ($a_{\text{super_freq}}$)

$$a_{\text{super_freq}} = \frac{F_{\text{super}} \cdot f_{\text{THz}} \cdot a_{\text{DPM}}}{E_{\text{vac,neb}} \cdot c}$$

With $F_{\text{super}} = 6.287 \times 10^{-19}$ N (magnetar magnetic force):

$$\boxed{a_{\text{super_freq}} = 1.048 \times 10^{-21} \text{ m/s}^2}$$

Term 5: Aether Resonance ($a_{\text{aether_res}}$)

$$a_{\text{aether_res}} = \frac{F_{\text{aether}} \cdot a_{\text{DPM}}}{E_{\text{vac,neb}}}$$

$$\boxed{a_{\text{aether_res}} = 3.900 \times 10^{-38} \text{ m/s}^2}$$

Term 6: Vacuum Concentration Reactivity ($U_{\text{g4i}} / E_{\text{react}}$ term)

$$U_{\text{g4i}} = f(E_{\text{react}}), \quad E_{\text{react}} = 1046 \cdot e^{-0.0005 \cdot t}$$

At $t = 3.799 \times 10^{10}$ s:

$$E_{\text{react}} = 1046 \cdot e^{-0.0005 \cdot 3.799 \times 10^{10}} \approx 0$$

$$\boxed{U_{\text{g4i}} \approx 0 \text{ m/s}^2 \quad \text{system too old – reacted to zero}}$$

Term 7: Quantum Frequency Coupling ($a_{\text{quantum_freq}}$)

$$a_{\text{quantum_freq}} = \frac{\hbar \cdot \Delta x \cdot \Delta p}{2\pi \cdot t_{\text{Hubble}}}$$

$$\boxed{a_{\text{quantum_freq}} = 1.708 \times 10^{-66} \text{ m/s}^2}$$

Term 8: Aether Frequency Coupling ($a_{\text{Aether_freq}}$)

$$a_{\text{Aether_freq}} = \frac{F_{\text{aether}} \cdot E_{\text{vac,neb}}}{m_{\text{eff}} \cdot c \cdot t_{\text{Hubble}}}$$

$$\boxed{a_{\text{Aether_freq}} = 1.863 \times 10^{-84} \text{ m/s}^2 \quad \text{(MINIMUM – weakest term)}}$$

Term 9: Fluid Frequency Coupling ($a_{\text{fluid_freq}}$) — DOMINANT

$$a_{\text{fluid_freq}} = f_{\text{fluid}} \cdot \frac{E_{\text{vac,neb}} \cdot V_{\text{sys}}}{E_{\text{vac,ISM}} \cdot c}$$

With $f_{\text{fluid}} = 1.269 \times 10^{-14}$ Hz for SGR1745:

$$a_{\text{fluid_freq}} = 1.773 \times 10^{-9} \text{ m/s}^2 \quad \textbf{(DOMINANT)}$$

Term 10: Oscillatory Term (Osc_term)

$$\text{Osc_term} = k_{\text{osc}} \cdot \cos(\pi t_n) \cdot \omega_{\text{osc}} \cdot t$$

At steady state (no active oscillation):

$$\text{Osc_term} = 0 \text{ m/s}^2$$

Term 11: Expansion Frequency (aexp_freq)

$$a_{\text{exp_freq}} = \frac{f_{\text{exp}}}{c} \cdot E_{\text{vac,neb}} \cdot a_{\text{DPM}} \cdot E_{\text{vac,ISM}} \cdot c$$

Where $f_{\text{exp}} = 2\pi \cdot H(z) \cdot t = 1.373 \times 10^{-8}$ Hz:

$$a_{\text{exp_freq}} = 1.623 \times 10^{-57} \text{ m/s}^2$$

Term 12: TRZ Neutrino Coupling (fTRZ)

$$f_{\text{TRZ}} = k_{\eta} \cdot E_{\text{vac,neb}} / (m_{\nu} \cdot c^2)$$

With $k_{\eta} = 10^{-113}$:

$$f_{\text{TRZ}} = 0.1 \text{ m/s}^2 \quad \text{(canonical parametric value)}$$

Note: fTRZ = 0.1 is a parametric coupling constant, not a computed acceleration. The tiny $k_{\eta} = 10^{-113}$ ensures TRZ coupling is negligible in standard physics but registers as a theoretical floor.

4. Complete Spectral Ladder Table (Ranked by Magnitude)

Rank	Term	Value (m/s ²)	Dynamic Range vs afluid
1 (DOMINANT)	afluid_freq	1.773×10 ⁻⁹	—
2	asuper_freq	1.048×10 ⁻²¹	×10 ⁻¹²
3	aaether_res	3.900×10 ⁻³	×10 ⁻²
4	aTHz	1.182×10 ⁻³³	×10 ⁻²
5	aDPM	3.545×10 ⁻²	×10 ⁻³³
6	avac_diff	3.545×10 ⁻³	×10 ⁻
7	aexp_freq	1.623×10 ⁻	×10 ⁻
8	aquantum_freq	1.708×10 ⁻	×10 ⁻
9 (MINIMUM)	aAether_freq	1.863×10 ⁻	×10 ⁻
— (zero)	Ug4i	≈ 0	(ancient system)
— (zero)	Osc_term	0	(no active osc.)
— (param)	fTRZ	0.1	(coupling constant)

Total resonance MUGE for SGR1745 ≈ 1.773×10⁻⁹ m/s² (fluid-dominated; all other terms negligible in sum but physically distinct mechanisms)

5. Dynamic Range Analysis

78-order dynamic range from $a_{\text{fluid_freq}} = 1.773 \times 10^{-9}$ to $a_{\text{Aether_freq}} = 1.863 \times 10^{-84}$.

This 78-order span reflects the multi-scale physics encoded in UQFF:

Fluid scale (10^{-12}): Macroscopic vacuum coupling to system volume — dominant at magnetar scale

Super/Aether scale (10^{-21} to 10^{-3}): Magnetic quantum resonances

THz/DPM scale (10^{-33} to 10^{-2}): Dipole phase modulation regime

Quantum/Cosmic scale (10^{-100} to 10^{-120}): Hubble-epoch vacuum fluctuations

The terms **span from stellar surface gravity to quantum vacuum fluctuations** — exactly what a Unified Quantum Field Framework should encode.

6. Term Hierarchy Law (SGR1745 Compact Object)

$a_{\text{fluid}} \gg a_{\text{super}} \gg a_{\text{aether_res}} \gg a_{\text{THz}} \gg a_{\text{DPM}} \gg a_{\text{vac}} \gg a_{\text{exp}} \gg a_{\text{quantum}} \gg a_{\text{Aether}}$

For a compact magnetar at $r = 10^4$ m, $V_{\text{sys}} = 4.189 \times 10^{12} \text{ m}^3$, the dominant mechanism is the **vacuum energy density coupling through the system volume**: the product $E_{\text{vac,neb}} \cdot V_{\text{sys}} / (E_{\text{vac,ISM}} \cdot c)$ sets the scale of gravitational coupling for compact objects.

7. Unit Test Canonical Reference Values

All 12 term values confirmed via C++ unit test suite in `grok_share_11254865.txt` (lines ~7000-7600):

```
test_compute_aDPM() → expected: 3.545e-42 m/s² ✓
test_compute_aTHz() → expected: 1.182e-33 m/s² ✓
test_compute_avac_diff() → expected: 3.545e-53 m/s² ✓
test_compute_asuper() → expected: 1.048e-21 m/s² ✓
test_compute_aaether() → expected: 3.900e-38 m/s² ✓
test_compute_Ug4i() → expected: ~0.0 m/s² ✓
test_compute_aquantum() → expected: 1.708e-66 m/s² ✓
test_compute_aAether() → expected: 1.863e-84 m/s² ✓
test_compute_afluid() → expected: 1.773e-9 m/s² ✓
test_compute_Osc_term() → expected: 0.0 m/s² ✓
test_compute_aexp() → expected: 1.623e-57 m/s² ✓
test_resonance_MUGE() → expected: ~1.773e-9 m/s² ✓
```

8. References Within Codebase

PAPER_371: MUGE 12-Term Superconductive Resonance — framework definition

PAPER_383: Ug4i Transient Age-Dependent Decay Law — explanation for $Ug4i=0$

PAPER_384: Sagittarius A* full decomposition — comparison between systems

MUGESuperconductive12TermResonanceCalculator (CP4 #14): Full implementation

Source: *grokshare11254865.txt* lines ~2920–2950 + unit tests ~7000–7600 | Session 104 | First per-term numeric tabulation of all 12 resonance MUGE terms for any system